



III. Solutions and Answers

The suggested solutions and answers below are only for your reference and have not been verified to be error-free. If you have any doubt, please discuss with your tutor to clarify.

MCQs

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
D	D	B	B	B	C	C	D	B	A	A	B	C	A	D

16	17	18	19	20	21	22	23	24	25	26	27			
D	A	B	B	D	C	D	C	B	A	D	D			

* A.C. transformer

- D** The magnetic field at a distance r from a long straight wire carrying a current I is $B = \frac{\mu_0 I}{2\pi r}$.
As the wire moves towards the flat coil, the magnetic flux linkage of the coil increases at an increasing rate. According to Faraday's law, the induced e.m.f. increases as the wire approaches the coil. Thus, current increases.
- D** As the current in the solenoid varies sinusoidally, the amplitude of the induced e.m.f. in the search coil is proportional to the rate of change of magnetic flux linkage, which also varies sinusoidally.
Along the solenoid, the magnetic flux density has approximately uniform but reduces to half towards the ends of the solenoid.
Hence, amplitude of the induced e.m.f. in the search coil is almost constant along the length of the solenoid and half at the ends.

- B** Average induced e.m.f.

$$\langle E \rangle = \frac{\Delta \Phi}{\Delta t} = \frac{160 \times 0.40 \times 0.20 - (-160 \times 0.40 \times 0.20)}{2.0} = 12.8 \text{ V}$$

Average power generated in coil

$$\langle P \rangle = \frac{\langle E \rangle^2}{R} = \frac{12.8^2}{16} = 10 \text{ W}$$

- B** area perpendicular to the magnetic field $A_{\perp} = \pi r^2 \cos 45^\circ$

Average induced e.m.f.

$$\langle E \rangle = \frac{\Delta \Phi}{\Delta t} = \frac{BA_{\perp}}{\Delta t} = \frac{76 \times 10^{-3} \times \pi \times 0.037^2 \cos 45^\circ}{4.5 \times 10^{-3}} = 0.051 \text{ V}$$

- 5 B Change in magnetic flux linkage

$$\Delta\Phi = |0 - NBA \sin 60^\circ| = |0 - 3 \times 0.045 \times 0.030 \sin 60^\circ| = 0.035 \text{ Wb}$$

- 6 C The rotating magnetic field cuts the copper disc. According to Lenz's law, to 'oppose' the "change in flux linkage", the disc will rotate in the same direction as the magnet.
- 7 C Apply Fleming's right-hand rule, current flows from P to Q when the wire is travelling vertically downwards (perpendicular to horizontal magnetic field pointing into the page).
- 8 D When the bar magnet enters the solenoid, the flux linkage through the solenoid increases, inducing an e.m.f. in the coil.
When the bar magnet leaves the solenoid, the flux linkage through the solenoid decreases, inducing an e.m.f. in the opposite direction.
Since the magnet accelerates under gravity, the magnitude of the induced e.m.f. is larger when the magnet leaves the solenoid and the time spent in second half of the solenoid is shorter.
- 9 B According to Faraday's law, induced e.m.f. for a rotating coil is $E = NBA\omega \sin \omega t$.
So induced e.m.f. does not depend on resistance of coil.
- 10 A According to Lenz's law, a North pole will be induced on the left side of the coil when the North pole is pushed into the left end; a South pole will be induced on the right side when the South pole is pushed into the right end. Hence, direction of the induced current in the coil is the same for both cases.
Since speed is halved, rate of change of magnetic flux linkage in the coil is lower and hence current is reduced.
- 11 A According to Fleming's RHR, induced current would flow (if a closed circuit is present) from Q to P. Hence P is at higher potential (as PQ is an e.m.f. source).
Since the magnetic field is weaker further away from the wire, the rate of cutting of flux by the conductor is lower. Hence magnitude of induced e.m.f. is reduced.
- 12 B Area of triangles $A = L^2 \sin \theta$.
Average e.m.f. induced

$$\langle E \rangle = \frac{\Delta\Phi}{\Delta t} = \frac{BA - 0}{\Delta t} = \frac{BL^2 \sin \theta}{\Delta t} = \frac{0.30 \times 1.5^2 \sin 60^\circ}{0.10} = 5.85 \text{ V}$$
- 13 C Induced e.m.f. $E = NBA\omega \sin \omega t$
Maximum induced e.m.f.

$$E_0 = NBA\omega = 1 \times 1.2 \times \pi \times 0.20^2 \times 2\pi \times \frac{50}{60} = 0.790 \text{ V}$$
- 14 A By Lenz's law, the induced current in the small coil flows in a clockwise direction to cause attraction between the coils to oppose the decreasing flux linkage of the small coil.

- 15 D** According to Lenz's law, the direction of the induced e.m.f. would drives the induced current to oppose the fall of the copper loop. As such, there is an upward magnetic force acting on the loop, resulting in lower acceleration ($a < g$) throughout the fall.

- 16 D** For a rotating coil, maximum induced e.m.f.

$$E_0 = NBA\omega \propto B\omega$$

$$E'_0 = \frac{B'\omega'}{B\omega} \cdot E_0 = \frac{1}{3} \cdot \frac{2}{1} \cdot E_0 = \frac{2}{3} E_0$$

- 17 A** Induced e.m.f.

$$E = BAf = B \times \pi r^2 \times \frac{v}{2\pi r} = \frac{1}{2} Brv = \frac{1}{4} Bdv$$

- 18 B** Induced e.m.f.

$$|E| = \frac{d\phi}{dt} = B \times \frac{dA}{dt} = 1.2 \times \frac{6 \times 10^{-3}}{3} = 2.4 \times 10^{-3} \text{ T}$$

- 19 B** The magnetic flux density due to long solenoid

$$B = \mu_0 nI = \frac{\mu_0 NI}{L}$$

Induced e.m.f. between the rim and center of disk

$$|E| = \frac{d\Phi}{dt} = B \times \frac{dA}{dt} = \frac{\mu_0 NI}{L} \times \frac{A}{T} = \frac{\mu_0 NI}{L} \times \frac{A\omega}{2\pi} = \frac{0.16\mu_0 NIA\omega}{L}$$

Since the e.m.f. across the solenoid is 10 times larger, $E_s = \frac{1.6\mu_0 NIA\omega}{L}$.

- 20 D** The magnetic flux density B due to long solenoid

$$B = \mu_0 nI = \mu_0 nI_0 \sin^2 \omega t$$

Induced e.m.f. in coil

$$E = -\frac{d\Phi}{dt} = -A \times \frac{dB}{dt} = -A\mu_0 nI_0 \omega \sin(2\omega t) = -E_0 \sin(2\omega t)$$

- 21 C** By Fleming's right-hand rule, induced current flows from X to Y along the rod. So, Y is at a higher potential. Due to the induced current, the rod experiences a magnetic force (using Fleming's left-hand rule) towards the left.

- 22 D** Root-mean-square voltage in secondary coil

$$V_s = \frac{N_s}{N_p} \times V_p = 2 \times 120 = 240 \text{ V}$$

Current through resistor

$$I_R = \frac{V_s}{R} = \frac{240}{8.0} = 30 \text{ A}$$

Since transformer is 75% efficient

$$0.75I_p V_p = I_s V_s$$

$$I_p = \frac{I_s V_s}{0.75 V_p} = \frac{30 \times 240}{0.75 \times 120} = 80 \text{ A}$$

- 23 C** Magnetic flux in the core of the transformer varies sinusoidally.

Hence the induced e.m.f. in the secondary coil also varies sinusoidally ($E = -d\Phi/dt \propto dI/dt$). Hence secondary e.m.f. is greatest at $t = 0, \frac{1}{2}T, T, \dots$, i.e., $E_s = V_0 \cos \omega t$.

Hence power output at load

$$P \propto \cos^2 \omega t$$

- 24 B** Since the supply voltage is halved, the power dissipated by the bulb is 9 W.

Assuming the transformer to be 100% efficient and $P_s = I_{\text{rms}} V_{\text{rms}} = 9 \text{ W}$,

$$I_{\text{rms}} = \frac{9}{\frac{1}{\sqrt{2}} \times 1.4} = 9.1 \text{ A}$$

- 25 A** Power delivered to load

$$P_{\text{load}} = I_s^2 R = 1.5^2 \times 10 = 22.5 \text{ W}$$

Assuming transformer is 100% efficient, then

$$P_{\text{supply}} = I_p V_p = 22.5 \text{ W} \quad V_p = 4.5 \text{ V}$$

- 26 D** Minimum power supplied (since transformer is not 100% efficient)

$$P = I_{\text{rms}} V_{\text{rms}} > 4 \times 24 = 96 \text{ W}$$

Minimum current from power supply

$$I_{\text{rms}} > \frac{96}{240} = 0.40 \text{ A}$$

- 27 D** Maximum power supplied assuming 100% transformer efficiency,

$$P_{\text{max}} = I_p V_p = 10 \times 230 \times 10^3 = 2.3 \times 10^6 \text{ W}$$

Maximum number of streetlamps connected in parallel

$$N = \frac{P_{\text{max}}}{P_{\text{lamp}}} = \frac{2.3 \times 10^6}{150} = 15000$$

Structured Questions

1. [ACJC/H2/Prelims 2018/P3/6 Part]

- c. i. That area represents the decrease in magnetic flux linkage of the coil when the magnet leaves the coil after passing through the centre of the coil (centre of magnet coincides with centre of coil).
- ii. As the magnet approaches the coil, the coil experiences an increasing magnetic flux linkage at an increasing rate. By Faraday's law, an increasing e.m.f. is induced in the coil.

As the magnet approaches the centre of the coil, the induced e.m.f. decreases to zero when the flux linkage is maximum.

As the magnet falls out of the coil, the coil experiences a decreasing magnetic flux linkage. Therefore, the induced e.m.f. has an opposite polarity but with a greater and narrower peak.

Note: The areas under graph are equal.

- iii. With a higher entry speed, the rate of change of magnetic flux linkage of the coil is higher. Hence the both peaks are correspondingly higher.

The duration of each peak is shorter as the magnet passes through the coil in a shorter time.

Note: The changes in magnetic flux linkage (area under V - t graph) when entering and leaving the coil are the same.

2. [AJC/H2/Prelims 2018/P3/6 Modified]

- a. From Fig. 2.2, at $x = 5.00$ cm, $B = 44$ mT.

Magnetic flux linkage

$$\Phi = NBA = 180 \times 44 \times 10^{-3} \times \pi \times \left(5.7 \times 10^{-3} / 2\right)^2 = 2.02 \times 10^{-4} = 2.0 \times 10^{-4} \text{ Wb}$$

- b. i. From the graph,

$$|\Delta B| = 44 - 5 = 39 \text{ mT}$$

Average induced e.m.f.

$$\langle E \rangle = \frac{\Delta \Phi}{\Delta t} = NA \left| \frac{\Delta B}{\Delta t} \right| = 180 \times \pi \times \left(5.7 \times 10^{-3} / 2\right)^2 \times \frac{39 \times 10^{-3}}{0.30} = 5.97 \times 10^{-4} = 6.0 \times 10^{-4} \text{ V}$$

Acceptable range for $\langle E \rangle$ is from 5.8×10^{-4} V to 6.0×10^{-4} V.

- ii. Since the magnetic flux density though the coil decreases at a decreasing rate with distance, to keep the induced e.m.f. constant, the speed of the coil must increase with distance x .

Note: $\frac{d\Phi}{dt} = A \cdot \frac{dB}{dt} = A \cdot \frac{dB}{dx} \cdot \frac{dx}{dt}$

- c. As the coil moves away from the magnet, the decreasing magnetic flux linkage of the coil produced an induced e.m.f. in the coil according to Faraday's law.

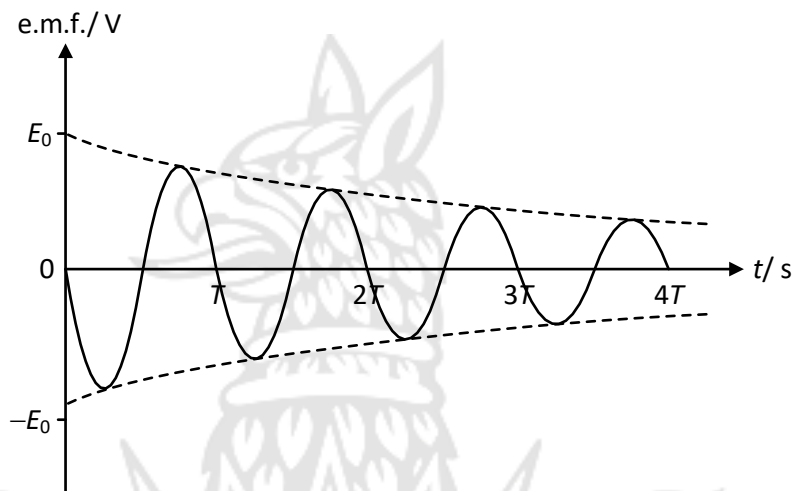
This induced e.m.f. drives a current in the closed circuit of finite resistance.

This electrical energy (current) is dissipated as heat due to Joule heating.

3. [CJC/H2/Prelims 2018/P2/7]

- a. i. While the frame is in motion, its magnetic flux linkage changes (or the wire ST cuts the magnetic field lines). According to Faraday's law, e.m.f. induced is proportional to the rate of change of magnetic flux linkage (or rate of cutting of magnetic field lines by wire ST). Hence, an e.m.f. is induced in the wire ST.
- ii. Since the wire frame is undergoing simple harmonic motion, its velocity varies sinusoidally with time. So the magnetic flux linkage of the frame and its rate of change vary sinusoidally. Hence the induced e.m.f. will vary sinusoidally with time.

b.



With a closed circuit, the induced e.m.f. drives an induced current that tends to oppose the changes that causes it (Lenz's law). The current causes a magnetic force to act on the wire ST to oppose the motion of the frame.

Hence, the oscillation is lightly damped and the induced e.m.f. decreases.

Note: The period of oscillations is slightly longer than for free oscillations.

- ii. When the frame is completely in the magnetic field, there is no change in magnetic flux linkage of the frame while it oscillates. So no e.m.f. is induced in the frame.

The graph is a horizontal line with at e.m.f. at zero.

4. [EJC/H2/Prelims 2018/P3/5]

- a. Faraday's law of electromagnetic induction states that the induced e.m.f. is proportional to the rate of change of magnetic flux linkage.
- b. The seismic waves cause the case and the coil (attached to the case) to move relative to the magnet. This results in a changing magnetic flux linkage of the coil.
By Faraday's law, an e.m.f. is induced across the coil.

- c. Induced e.m.f.

$$E = \frac{d\Phi}{dt} = N \frac{d\phi}{dt}$$

Rate of change of magnetic flux

$$\frac{d\phi}{dt} = \frac{E}{N} = \frac{85 \times 10^{-3}}{50} = 1.7 \times 10^{-3} \text{ V or Wb s}^{-1}$$

- d.
- Magnet with stronger magnetic field
 - Coil with more turns
 - Spring with a lower spring constant

5. [HCI/H2/Prelims 2018/P3/6]

a.	Incorrect statement	Correct number or unit
	The weight of a person is about <u>700 kg</u> .	700 N
	The atmospheric pressure at sea level is about <u>$1.0 \times 10^5 \text{ N m}^2$</u> .	$1.0 \times 10^5 \text{ N m}^{-2}$
	1 GW is <u>10</u> times higher than 1 MW.	1000

- b. The advancement in Science will require better precision at the realm of the very small objects. The physical sample has its associate uncertainty in mass. Any mass measuring instrument calibrated from it will inherit this uncertainty. When that mass measuring instrument is measuring very small masses, the percentage uncertainty in mass will increase with the decreasing mass measured.

- c. i. To exert an anti-clockwise torque, the magnetic force must be directed downward.
Using Fleming's left-hand rule, the current flows from left to right through the conductor.
- ii. 1. The "free" or delocalised electrons within the rod move upward with the rod. Due to their motion in the magnetic field, these electrons experience a magnetic force directed to the left which causes an accumulation of net negative charge on the left and leaving a net positive charge on the right. This is the potential difference (induced e.m.f.) across the ends of the conductor.

OR

As the rod moves upwards, it cuts the magnetic flux. By Faraday's law, an e.m.f. (proportional to the rate of cutting of magnetic flux) will be induced across the ends of the rod.

2. At equilibrium, the magnetic force on the electron is balanced by the electric force due to the internal electric field set up at the two ends.

$$F_B = F_E$$

$$\Rightarrow Beu = eE = e \frac{V}{L}$$

$$\Rightarrow V = BLu$$

Alternatively, according to Faraday's law

$$|V| = \frac{d\Phi}{dt} = \frac{d(BLx)}{dt} = BLu$$

where x is the vertical displacement of conductor at time t .

- *iii. From c.i. and c.ii.,

$$m = \frac{BIL}{g} = \frac{VI}{ug}$$

Since $V = \frac{1}{2} hC/e$ and $I = \frac{1}{2} kne$,

$$m = \frac{VI}{ug} = \frac{(\frac{1}{2} hC/e) \times \frac{1}{2} kne}{ug} = \frac{knC}{4ug} \cdot h$$

6. [HCI/H2/Prelims 2018/P3/7 Modified]

- a. The weber is the magnetic flux through an area of 1 m^2 when a magnetic flux density of 1 T is directed perpendicularly to that area.
- b. i. The magnetic flux linkage of the frame increases linearly as the frame ABCD enters the field. According to Faraday's law, an e.m.f. is induced along AB.

By Lenz's Law, the induced e.m.f. drives a current in the frame in an anti-clockwise direction which in turn produces a magnetic force on AB directed to the left.

OR

As the side AB enters the magnetic field, the delocalised electrons experience magnetic forces that accelerate them in the clockwise direction around the frame. This causes a current to flow in an anti-clockwise direction which in turn produces a magnetic force on AB directed to the left.

- ii. Magnetic flux linkage of frame

$$\Phi = BA = Bwx$$

where x is the length of the frame in the magnetic field.

Magnitude of induced e.m.f.

$$|E| = \frac{d\Phi}{dt} = \frac{dBwx}{dt} = Bwv$$

- iii. Induced current in frame

$$I = \frac{E}{R} = \frac{Bwv}{R}$$

Magnetic force on frame (AB)

$$F = BIl = \frac{B^2 w^2 v}{R}$$

Acceleration of frame

$$a = \frac{F}{m} = \frac{B^2 w^2 v}{mR} \quad (\text{directed to left})$$

- c. i. The area under graph represents $2L$. Using trapezium rule,

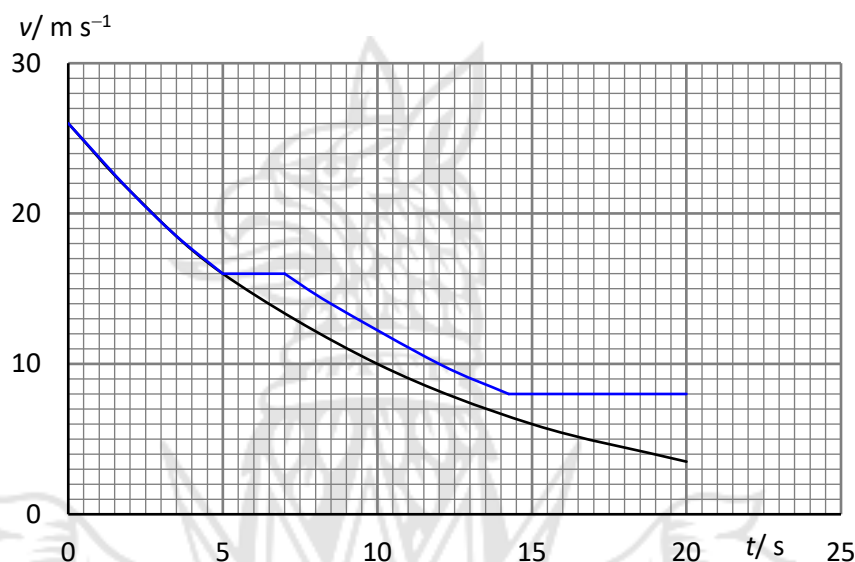
$$2L = \frac{1}{2}(26 + 2 \times 16 + 2 \times 10 + 2 \times 6 + 3.5) \times 5 = 234 \text{ m}$$

$$\Rightarrow L = 117 \text{ m}$$

Acceptable range for L is 110 m to 125 m.

- ii. When the frame is entirely in the magnetic field, there is no change in magnetic flux linkage so no e.m.f. or current is induced in the frame and the train experiences no retarding force. Therefore, increasing the length PQ does not change the exit speed.

iii.



1. When AB enters the magnetic field, v decreases at the same rate as before.
2. AB exits the magnetic field but CD has yet to enter the magnetic field. No change in flux linkage, no induced e.m.f. and current. No retarding force. So, v is constant.
3. When CD enters the magnetic field, v continues to decrease at same rate as before.
4. Finally, the train leaves the magnetic field at a higher speed as the overall time under deceleration is shorter.

- iv. 1. Multiple regions of magnetic fields that can be activated when malfunction occurs.
2. Use of electromagnets where stronger magnetic fields can be generated by increasing the current when malfunction occurs.

Note: Any modifications that relate to the dimensions of the metal frame are not accepted as these cannot be modified during a malfunction.

7. [IJC/H2/Prelims 2018/P3/9 Modified]

- a. i. Magnetic flux is the product of area A and the component of the magnetic flux density perpendicular to that area.

$$\phi = BA \cos \theta$$

- ii. Faraday's law of electromagnetic induction states that the induced e.m.f. is proportional to the rate of change of magnetic flux linkage.

Lenz's law states that the direction of induced e.m.f. is such as to cause effects to oppose the change producing it.

- b. i. Magnetic flux linkage at 30.0 mm position

$$\Phi_i = NBA = 50 \times 1.0 \times 10^{-3} \times 16.0 \times 10^{-4} = 8.0 \times 10^{-5} \text{ Wb}$$

At 10.0 mm position

$$\Phi_f = 50 \times 31.0 \times 10^{-3} \times 16.0 \times 10^{-4} = 2.48 \times 10^{-3} \text{ Wb}$$

- ii. Since $\langle E \rangle = \frac{\Delta \Phi}{\Delta t}$, time taken

$$\Delta t = \frac{\Delta \Phi}{\langle E \rangle} = \frac{(2.48 - 0.080) \times 10^{-3}}{0.80 \times 10^{-3}} = 3.0 \text{ s}$$

Average speed

$$\langle v \rangle = \frac{\Delta s}{\Delta t} = \frac{20.0 \times 10^{-3}}{3.0} = 6.7 \times 10^{-3} \text{ m s}^{-1}$$

- iii. Induced e.m.f.

$$E = \frac{d\Phi}{dt} = \frac{dNBA}{dt} = NA \cdot \frac{dB}{dt} = NA \cdot \frac{dB}{dx} \cdot \frac{dx}{dt} = NA \cdot \frac{dB}{dx} \cdot v$$

To maintain a steady induced e.m.f. in the coil, the speed of the coil should vary inversely to dB/dx .

Since dB/dx increases as the coil moves nearer the bar magnet, the speed of the coil must decrease as it approaches the magnet.

- c. i. For a rotating coil, $E = NBA\omega \sin \omega t$

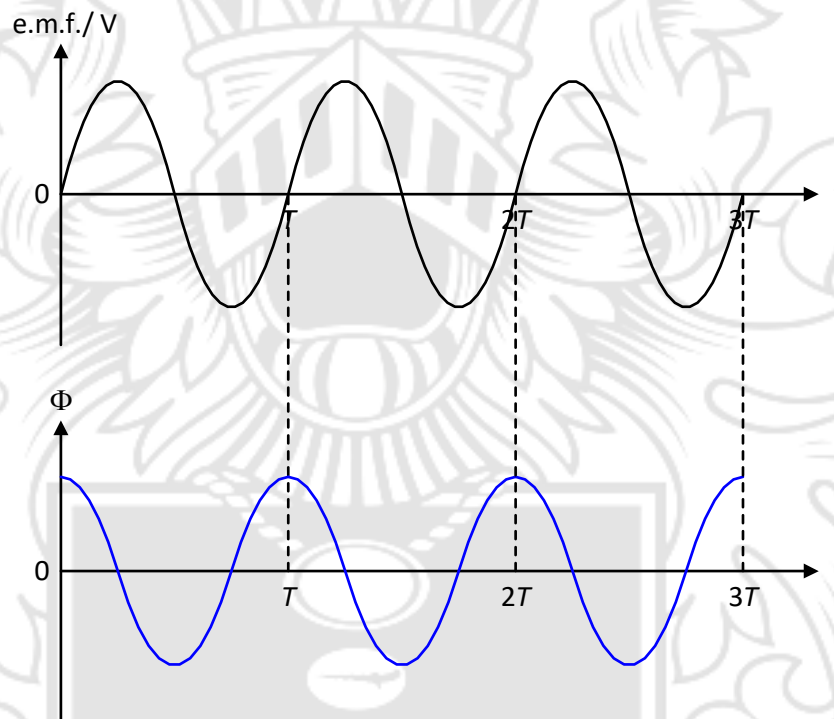
Maximum induced e.m.f.

$$E_0 = NBA\omega = 800 \times 0.50 \times 0.050 \times 0.080 \times 2\pi \times \frac{240}{60} = 40.2 \text{ V}$$

- ii. Faraday's law of electromagnetic induction

$$E = -\frac{d\Phi}{dt}$$

- iii.



- iv. Maximum current

$$I_0 = \frac{E_0}{R_T} = \frac{40.2}{0.60 + 11.4} = 3.35 \text{ A}$$

- v. 1. The term r.m.s. value refers to the square root of the mean squared values.
2. The r.m.s. current is the steady direct current which would dissipate electrical energy at the same rate in a given resistance as an alternating current.

- vi. r.m.s. current

$$I_{\text{rms}} = \frac{I_0}{\sqrt{2}} = \frac{3.35}{\sqrt{2}} = 2.37 \text{ A}$$

- vii Average power dissipated

$$\langle P \rangle = I_{\text{rms}}^2 R = 2.37^2 \times 11.4 = 64.0 \text{ W}$$

8. [JJC/H2/Prelims 2018/P3/9 Part Modified]

- c. Faraday's law of electromagnetic induction states that the induced e.m.f. is proportional to the rate of change of magnetic flux linkage.

Lenz's law states that the direction of induced e.m.f. is such as to cause effects to oppose the change producing it.

- d. i. Magnetic flux linkage $\Phi = NBA = 500 \times 2.0 \times 10^{-4} \times 0.050^2 \times \sin 35^\circ = 1.43 \times 10^{-4} \text{ Wb}$
- ii. 1. For a rotating coil, $E = NBA\omega \sin \omega t$
Maximum induced e.m.f.

$$E_0 = NBA\omega = 500 \times 2.0 \times 10^{-4} \times 0.050^2 \times 2\pi \times 100 = 0.157 \text{ V}$$

2. Average power generated

$$\langle P_{AC} \rangle = \frac{V_0^2}{2R} = \frac{0.157^2}{2 \times 5.00} = 2.47 \times 10^{-3} \text{ W}$$

3. Average power generated of half-wave rectified voltage

$$\langle P_{\text{rectified}} \rangle = \frac{1}{2} \langle P_{AC} \rangle = 1.23 \times 10^{-3} \text{ W}$$

9. [MI/H2/Prelims 2018/P3/8 Modified]

- a. i. Magnetic flux is the product of area A and the component of the magnetic flux density perpendicular to that area $B \cos \theta$.

$$\phi = BA \cos \theta$$

- ii. Magnetic flux through coil

$$\phi = BA = 25 \times 10^{-3} \times 0.150 \times 0.060 = 2.25 \times 10^{-4} \text{ Wb}$$

- b. i. As the magnetic flux density through the coil varying sinusoidally with time, the magnetic flux linkage of the coil is also varying sinusoidally with time.

By Faraday's law, an e.m.f. is induced in the coil.

Since the induced e.m.f. is proportional to the rate of change of magnetic flux linkage through the coil, it is

$$\propto \frac{d\phi}{dt} = A \frac{dB}{dt} = A \frac{d}{dt} (B_0 \sin \omega t) \propto \cos \omega t. \text{ Hence, it varies sinusoidally.}$$

- ii. Magnetic flux linkage

$$\Phi = NB_0 A \cos \omega t$$

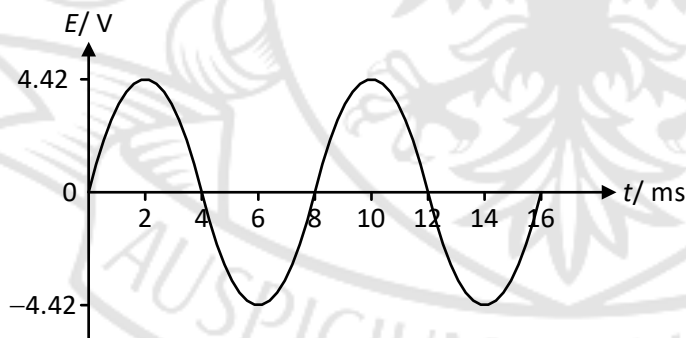
For a rotating coil, $E = NBA\omega \sin \omega t$

$$E = -\frac{d\Phi}{dt} = -\frac{d(NB_0 A \cos \omega t)}{dt} = -NB_0 A \frac{d(\cos \omega t)}{dt} = NB_0 A \omega \sin \omega t = NB_0 A \cdot \frac{2\pi}{T} \cdot \sin\left(\frac{2\pi}{T} \cdot t\right)$$

- iii. Maximum induced e.m.f.

$$E_0 = NB_0 A \omega = 25 \times 25 \times 10^{-3} \times 0.150 \times 0.060 \times \left(\frac{2\pi}{8.0 \times 10^{-3}}\right) = 4.42 \text{ V}$$

- iv.



- c. i. Root-mean-square voltage of input

$$V_p = \frac{V_0}{\sqrt{2}} = \frac{4.42}{\sqrt{2}}$$

Number of turns in secondary coil

$$N_s = \frac{V_s}{V_p} \times N_p = \frac{240}{4.42/\sqrt{2}} \times 15 = 1150$$

ii. With an efficiency of 81%,

$$0.81 \times \text{power input} = \text{power output}$$

$$0.81 \times I \times 12 = 0.25 \times 240$$

$$I = 6.17 \text{ A}$$

d. i. Voltages can be step up (for efficient transmission) and step down (for consumer use) efficiently using a transformer.

ii. Transmission current is lower when transmission voltage is higher for the same power.

A lower transmission current results in lower power loss due to Joules heating in the transmission cable.

10. [MJC/H2/Prelims 2018/P2/5]

a. Magnetic flux is the product of area A and the component of the magnetic flux density perpendicular to that area $B \cos \theta$.

$$\phi = BA \cos \theta$$

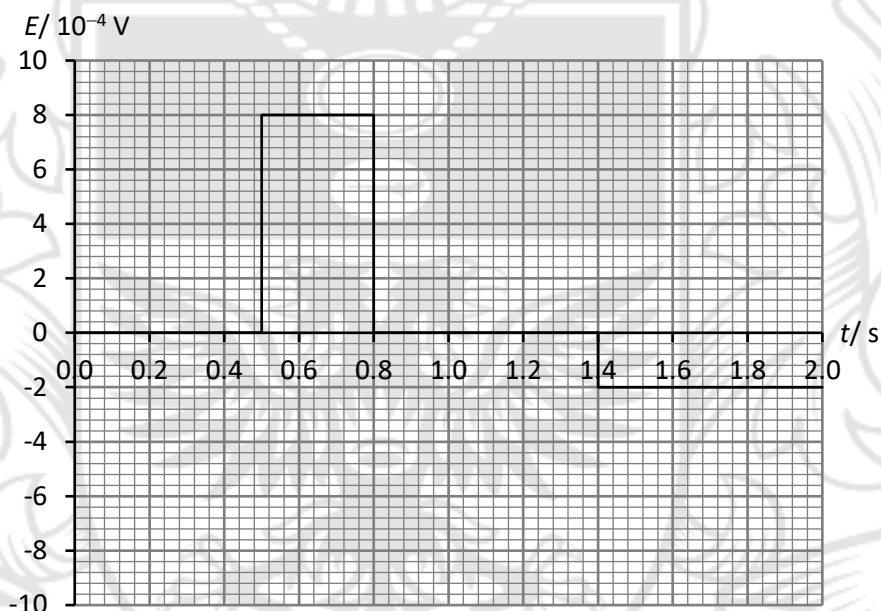
b. i. Magnetic flux linkage of coil Y

$$\Phi = NBA = N \times \mu_0 n I \times A = 80 \times 4\pi \times 10^{-7} \times \frac{400}{1.6} \times 3.8 \times \pi \times 0.020^2 = 1.2 \times 10^{-4} \text{ Wb}$$

ii. Mean induced e.m.f.

$$\langle E \rangle = \left| -\frac{\Delta \Phi}{\Delta t} \right| = \frac{2 \times 1.2 \times 10^{-4}}{0.30} = 8.0 \times 10^{-4} \text{ V}$$

iii.



$$E_1 = 0.0 \text{ V from } 0 \text{ s} \leq t \leq 0.5 \text{ s and } 0.8 \text{ s} \leq t \leq 1.4 \text{ s}$$

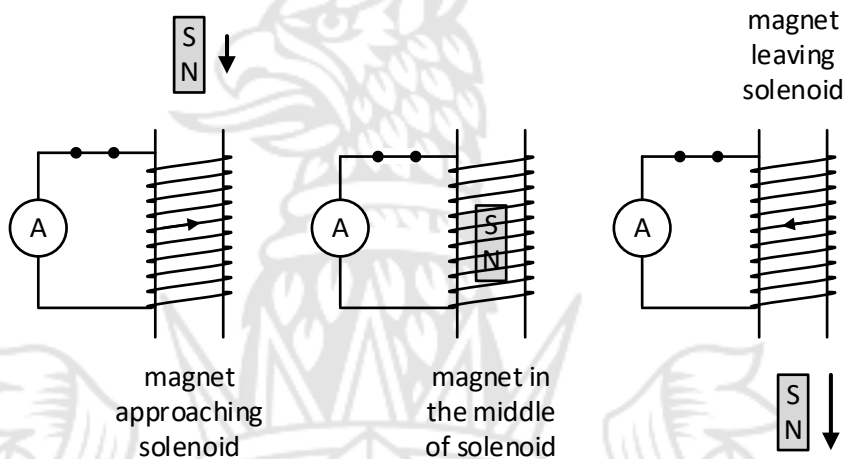
$$E_2 = 8.0 \times 10^{-4} \text{ V from } 0.5 \text{ s} \leq t \leq 0.8 \text{ s}$$

$$E_3 = -2.0 \times 10^{-4} \text{ V from } 1.4 \text{ s} \leq t \leq 2.0 \text{ s}$$

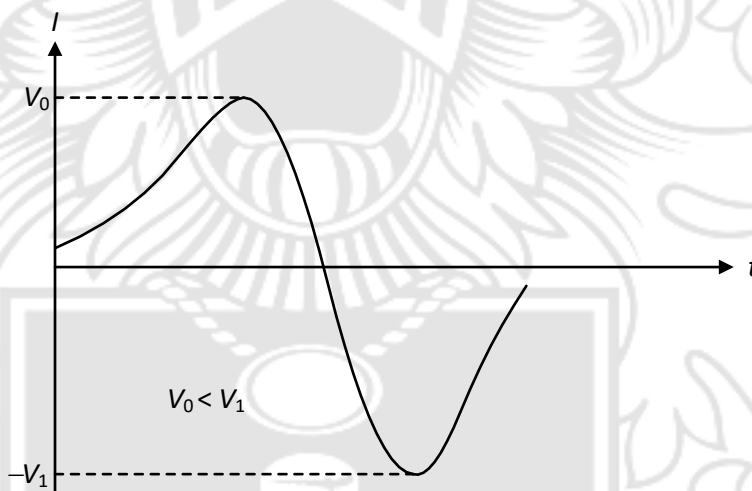
iv. The iron core increases the magnetic flux density in and around the solenoid, resulting in a greater rate of change of flux linkage of coil Y. Hence, the e.m.f. induced in coil Y is greater than in iii. (except when $E = 0.0 \text{ V}$).

11. [NYJC/H2/Prelims 2018/P3/6 Modified]

- a. Lenz's law states that the direction of induced e.m.f. is such as to cause effects to oppose the change producing it.
- b. i.

**Fig. 11.1a****Fig. 11.1b****Fig. 11.1c**

ii.



- iii. Since the switch is opened, there is no induced current in the solenoid to produce a magnetic field that opposes the motion of the magnet. Hence, the magnet leaves the solenoid with a greater speed.

12. [PJC/H2/Prelims 2018/P3/7]

- a. Magnetic flux is the product of area A and the component of the magnetic flux density perpendicular to that area $B \cos \theta$.

$$\phi = BA \cos \theta$$

- b. i. The suspended magnet is first repelled upwards as the falling magnet approaches the left coil.

As the left magnet approaches the left coil, there is a change in flux linkage associated with the left coil. By Faraday's law of EMI, there will be an induced e.m.f. in the left coil. By Lenz's law, this induced e.m.f. will cause an induced current to flow in such a direction as to oppose the change causing it, i.e. it will be such that the top end of the left coil will be a south pole to oppose the magnet's south pole from approaching. This direction of current flow will cause the top end of the right coil to become a south pole too, causing the suspended magnet to be repelled upwards.

As the dropped magnet leaves the left coil, the suspended magnet is attracted downwards.

As the dropped magnet leaves the left coil, the bottom end of the left coil becomes a south pole according to Lenz's law. This means the top ends of both coils become a North pole. An attraction of the suspended magnet downwards is observed.

- ii. With switch S opened, the suspended magnet will not move. Although there is an e.m.f. induced across the left coil, there is no induced current in the right coil and thus the right coil is not magnetised.

13. [RVHS/H2/Prelims 2018/P3/8]

- a. Faraday's law of electromagnetic induction states that the induced e.m.f. is proportional to the rate of change of magnetic flux linkage.
- b. i. As the magnet rotates, the magnetic field cutting the coils change in magnitude and direction, which cause the magnetic flux linkages in the coils to change.

With a constant angular speed, the magnetic flux linkage through each coil varies sinusoidally with time, i.e., $\Phi = NBA \cos \omega t$.

By Faraday's law, the induced e.m.f. in the coil, which is the rate of change of magnetic flux linkage, will also vary sinusoidally with time, i.e., $E = -NBA\omega \sin \omega t$.

- ii. Magnetic flux linkage

$$\Phi = NBA = 500 \times 0.80 \times \pi \times 0.050^2 = 3.14 \text{ Wb}$$

- iii. From the graph,

$$\Phi = NBA \cos\left(\frac{2\pi}{T} \cdot t\right) = 3.14 \cos\left(\frac{2\pi}{60/1800} \cdot t\right) = 3.14 \cos(60\pi \cdot t) \text{ Wb}$$

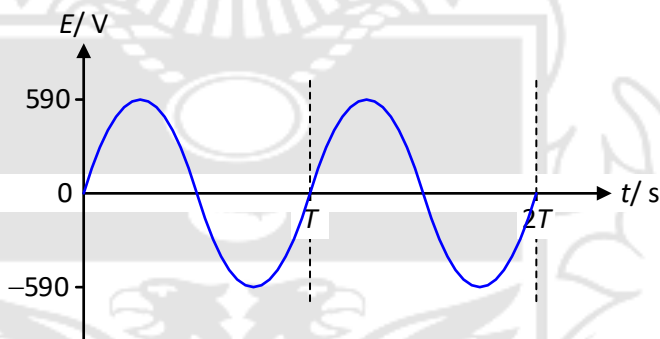
- iv. Induced e.m.f.

$$E = -\frac{d\Phi}{dt} = NBA\omega \sin \omega t$$

where the maximum induced e.m.f.

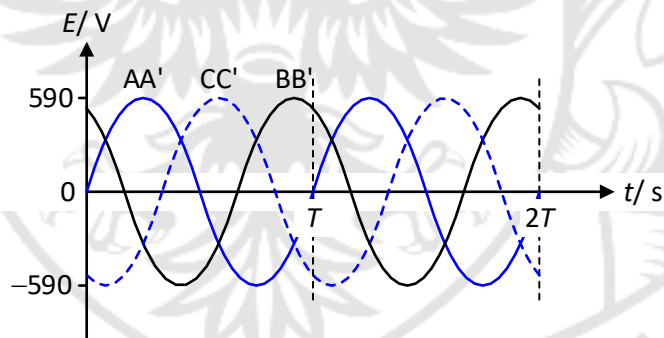
$$E_0 = NBA\omega = 3.14 \times 60\pi = 592 = 590 \text{ V}$$

- v.



Maximum induced e.m.f. occurs where the rate of change of the magnetic flux linkage is greatest at $\Phi = 0$.

- c.



The induced e.m.f. due to coil CC' and BB' are $1/3$ and $2/3$ cycle after that of AA'.

All induced e.m.f. have the same frequency and amplitude.

- d. i. Power transfer from the primary coil to the secondary coil is 100% efficient (no hysteresis loss in the iron core).
- ii. Peak secondary current

$$I_s = \frac{N_p}{N_s} \times I_p = \frac{30}{12000} \times 20 = 0.050 \text{ A}$$

Root-mean-square secondary current

$$\sqrt{\langle I_s \rangle} = \frac{I_s}{\sqrt{2}} = \frac{0.050}{\sqrt{2}} = 0.035 \text{ A}$$

- iii. Power transmission at higher voltage requires a lower transmission current.

A lower transmission current results in a lower power loss due to joules heating ($P_{\text{loss}} = I^2 R$) in the cable.

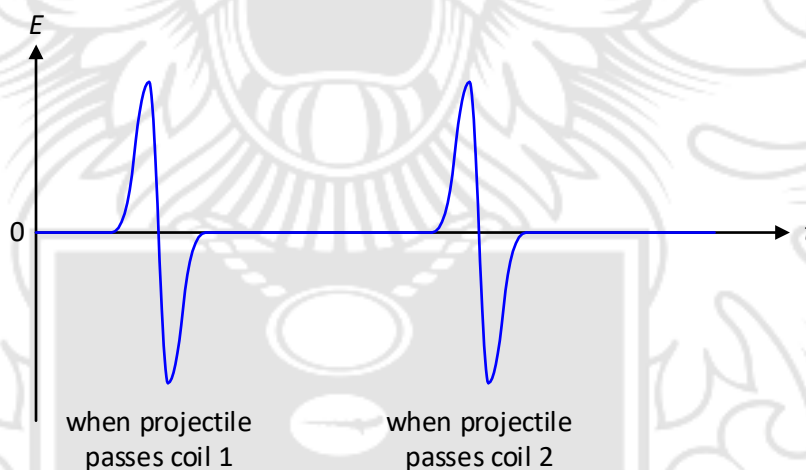
14. [VJC/H2/Prelims 2018/P2/5 Modified]

- a. Lenz's law states that the direction of induced e.m.f. is such as to cause effects to oppose the change producing it.

Connect a coil to a galvanometer. Move the north pole of the magnet into the coil and take note the direction of induced current flowing in the coil using a galvanometer. Using the right-hand grip rule, the direction of induced current is such that the induced magnetic field due is opposite to the magnetic field due to the magnet. This produces a magnetic force on the magnet to retard its motion.

- b. i. As the projectile approaches a coil, the magnetic field cutting the coil increases. So, the magnetic flux linkage of the coil increases. By Faraday's law, an e.m.f. is induced in the coil. When the projectile leaves the coil, the magnetic flux linkage of the coil decreases and an opposite e.m.f. is induced in the coil.

ii.



- iii. Speed of projectile

$$v = \frac{d}{t} = \frac{1.50}{2.40 \times 10^{-3}} = 625 \text{ m s}^{-1}$$

- iv. The pulses will be closer together as the bullet takes a shorter time to travel between the coils.

The induced e.m.f. of each pulse has a larger amplitude and a shorter duration as the coils experience a higher rate of change of magnetic flux linkage.

15. [YJC/H2/Prelims 2018/P3/6 Part]

- d. i. There is a change in magnetic flux linkage through each coil as the cart approaches and leaves the coil.
- ii. The direction of the induced e.m.f. is such that it drives an induced current to oppose the change producing it. As the cart approaches the coil, the magnetic flux linkage of the coil increases and an e.m.f. in one direction is induced. As the cart leaves the coil, the magnetic flux linkage decreases and an e.m.f. in the opposite direction is induced.
- iii. The cart accelerates down the slope with increasing speed. So, the increase in magnetic flux linkage in coil 2 occurs in a shorter time than in coil 1. Hence, the rate of change of magnetic flux linkage in coil 2 is greater, leading to a larger induced e.m.f. of pulse C.

16. [DHS/H2/Prelims 2018/P3/6]

a. Root-mean-square voltage

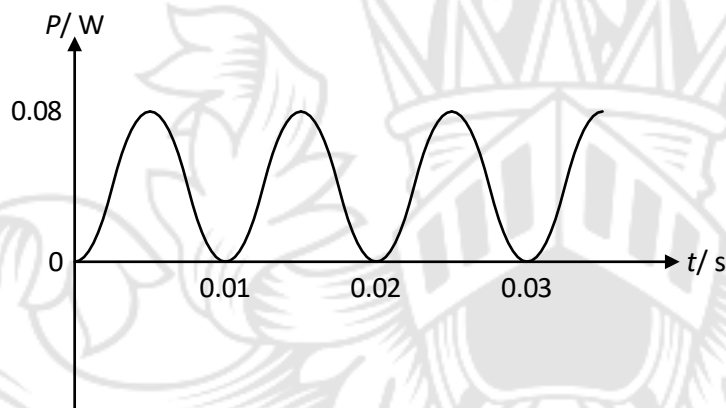
$$V_{\text{rms}} = \sqrt{\frac{2.0^2 \times 2 \times 10^{-3} + 1.0^2 \times 2 \times 10^{-3}}{10 \times 10^{-3}}} = 1.0 \text{ V}$$

b. Power transmission at higher voltages requires a lower transmission current since $P = IV$.
With lower transmission current, the power loss ($P = I^2 R$) through joule heating is lower.

c. i. Root-mean-square output voltage

$$V_s = \frac{N_s}{N_p} \times V_p = 71 \times 6.5 \times 10^{-3} = 0.46 \text{ V}$$

ii.



1. In forward biased direction, the diode has no resistance. Current flows clockwise through resistor R .
In reverse biased direction, diode has infinite resistance. No current flows through resistor R .
2. The p.d. across resistor R is half-wave rectified sinusoidal voltage.

17. [NYJC/H2/Prelims 2018/P2/8]

a. Ratio

$$\frac{N_p}{N_s} = \frac{V_p}{V_s} = \frac{12.0 \times 10^3}{240 \times 10^3} = 0.0500$$

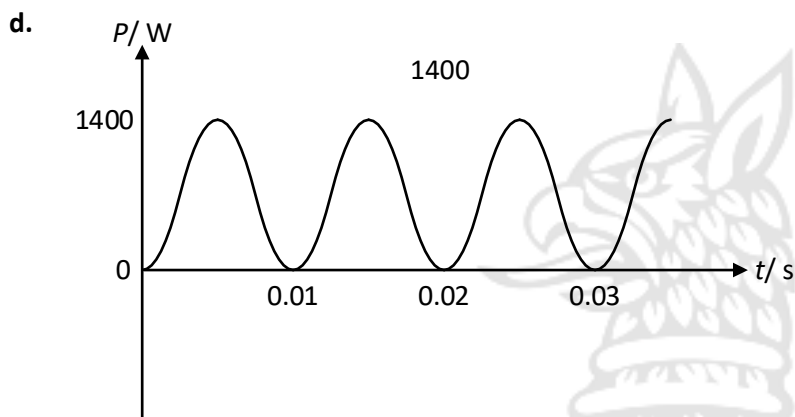
- b. i. The same power delivered at higher transmission voltage would result in a lower transmission current.
Lower transmission current results in lower power loss/dissipated in the transmission cables.
- ii. A transformer can easily step up (to lower transmission power loss) and step down (for safe consumer use) an alternative voltage.

c. i. Root-mean-square current

$$I_{\text{rms}} = \frac{\langle P \rangle}{V_{\text{rms}}} = \frac{700}{240} = 2.92 \text{ A}$$

ii. Peak output voltage

$$V_0 = \sqrt{2} V_{\text{rms}} = \sqrt{2} \times 240 = 339 \text{ V}$$



18. [SRJC/H2/Prelims 2018/P3/5]

- a. When an alternating current flows through the primary coil, it sets up an alternating magnetic field in the core which links the primary and secondary coils.

The secondary coil experiences alternating magnetic flux linkage due to the alternating magnetic field.

By Faraday's law of electromagnetic induction, the alternating flux linkage induces an alternating electromotive force in the secondary coil.

- b. i. Maximum magnetic flux density

$$B_{\max} = \mu_0 n I_0 = 4\pi \times 10^{-7} \times 900 \times \sqrt{2} \times 0.85 = 1.4 \times 10^{-3} \text{ T}$$

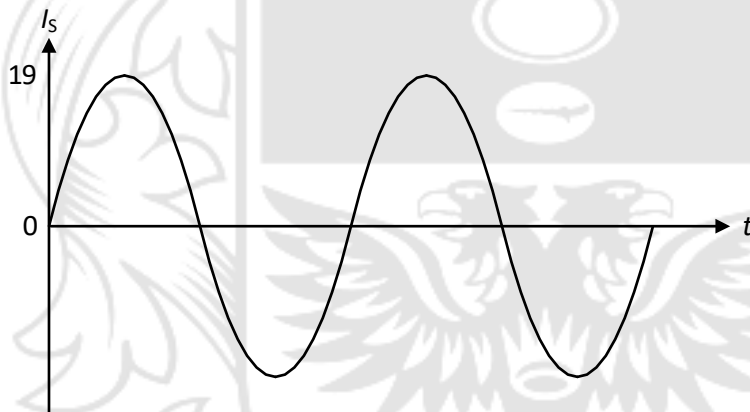
- ii. Root-mean-square current in secondary coil

$$I_s = \frac{N_p}{N_s} \times I_p = 16 I_p = 13.6 \text{ A}$$

Peak current in the load

$$I_0 = \sqrt{2} I_s = 19 \text{ A}$$

- iii.



- iv. Without the iron core, magnetic flux linking the secondary coil will decrease significantly due to flux leakage. The rate of change of magnetic flux linkage will also decrease.

By Faraday's law, the induced e.m.f. and hence the induced current in the secondary coil will decrease.

19. [DHS/H2/Prelims 2018/P3/6]

- a. Root-mean-square voltage

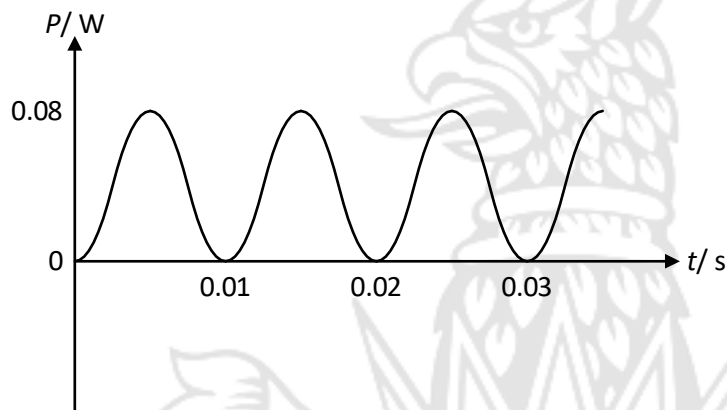
$$V_{\text{rms}} = \sqrt{\frac{2.0^2 \times 2 \times 10^{-3} + 1.0^2 \times 2 \times 10^{-3}}{10 \times 10^{-3}}} = 1.0 \text{ V}$$

- b. Power transmission at higher voltages requires a lower transmission current since $P = IV$. With lower transmission current, the power loss ($P = I^2 R$) through joule heating is lower.

- c. i. Root-mean-square output voltage

$$V_s = \frac{N_s}{N_p} \times V_p = 71 \times 6.5 \times 10^{-3} = 0.46 \text{ V}$$

ii.



- iii. 1. In forward biased direction, the diode has no resistance. Current flows clockwise through resistor R .
In reverse biased direction, diode has infinite resistance. No current flows through resistor R .
2. The p.d. across resistor R is half-wave rectified sinusoidal voltage.

20. [EJC/H2/Prelims 2018/P3/6]

- a. The root-mean-square current is that value of a steady direct current that would produce thermal energy at the same rate in a given resistor.
- b. i. Mean power $\langle P \rangle = I_{\text{rms}} V_{\text{rms}}$, so r.m.s. current

$$I_{\text{rms}} = \frac{\langle P \rangle}{V_{\text{rms}}} = \frac{1200}{120} = 10 \text{ A}$$

Peak current

$$I_0 = \sqrt{2} I_{\text{rms}} = 14 \text{ A}$$

- ii. Current $I = I_0 \sin \omega t = I_0 \sin(2\pi ft) = 14.1 \sin(100\pi t)$

- c. Mean total power in the primary coil

$$P_p = 1200 + 600 = 1800 \text{ W}$$

Since $P_p = I_p V_p$,

$$I_p = \frac{P}{V_p} = \frac{1800}{2.4 \times 10^3} = 0.75 \text{ A}$$

21. [NYJC/H2/Prelims 2018/P2/8]

- a. Ratio

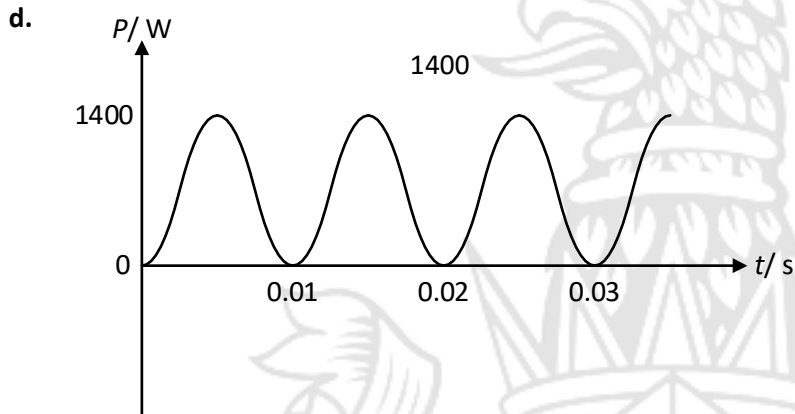
$$\frac{N_p}{N_s} = \frac{V_p}{V_s} = \frac{12.0 \times 10^3}{240 \times 10^3} = 0.0500$$

- b. i. The same power delivered at higher transmission voltage would result in a lower transmission current.
Lower transmission current results in lower power loss/dissipated in the transmission cables.
- ii. A transformer can easily step up (to lower transmission power loss) and step down (for safe consumer use) an alternative voltage.
- c. i. Root-mean-square current

$$I_{rms} = \frac{\langle P \rangle}{V_{rms}} = \frac{700}{240} = 2.92 \text{ A}$$

ii. Peak output voltage

$$V_0 = \sqrt{2}V_{rms} = \sqrt{2} \times 240 = 339 \text{ V}$$



22. [SRJC/H2/Prelims 2018/P3/5]

- a. When an alternating current flows through the primary coil, it sets up an alternating magnetic field in the core which links the primary and secondary coils.

The secondary coil experiences alternating magnetic flux linkage due to the alternating magnetic field.

By Faraday's law of electromagnetic induction, the alternating flux linkage induces an alternating electromotive force in the secondary coil.

- b. i. Maximum magnetic flux density

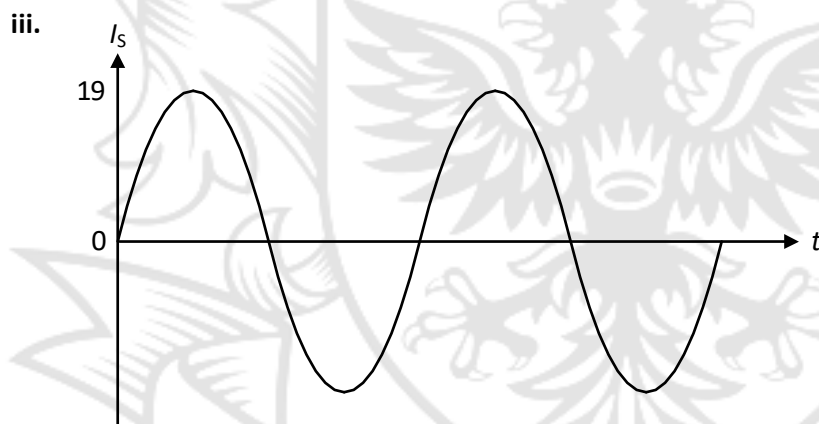
$$B_{max} = \mu_0 n I_0 = 4\pi \times 10^{-7} \times 900 \times \sqrt{2} \times 0.85 = 1.4 \times 10^{-3} \text{ T}$$

- ii. Root-mean-square current in secondary coil

$$I_s = \frac{N_p}{N_s} \times I_p = 16 I_p = 13.6 \text{ A}$$

Peak current in the load

$$I_0 = \sqrt{2} I_s = 19 \text{ A}$$



- iv. Without the iron core, magnetic flux linking the secondary coil will decrease significantly due to flux leakage. The rate of change of magnetic flux linkage will also decrease.

By Faraday's law, the induced e.m.f. and hence the induced current in the secondary coil will decrease.